

The Effect of State Antitakeover Laws on Corporate Tax Avoidance

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ABSTRACT

We examine the implications of enhanced managerial control afforded by state antitakeover laws for tax avoidance and find the strength of antitakeover statutes in a state is negatively related to tax avoidance of firms incorporated in that state. In testing underlying mechanisms, we find that the empirical results support the price discount concern hypothesis (i.e., managers refrain from tax avoidance activities to prevent potential stock price discounts induced by heightened investor perception of tax-related rent extraction in firms with severe agency problems). By contrast, the results are inconsistent with the quiet life hypothesis (i.e., managers entrenched by state antitakeover protection enjoy the quiet life by underinvesting in tax planning opportunities). Our findings are generally consistent with the takeover market phenomenon that one of managers' concerns lies in their inability to re-contract away the adverse consequences of perceived agency problems arising from state antitakeover protection.

Keywords: tax avoidance; managerial control; state antitakeover laws; agency problems; takeover market.

JEL Classifications: M41, M48, G34, G18.

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I. INTRODUCTION

Tax avoidance is an important corporate decision and has significant implications for firm value (Desai and Dharmapala 2009). Prior studies in the tax literature suggest that individual managers and their incentives have incremental influence on firms' tax avoidance beyond firm characteristics (Dyreng et al. 2010; Rego and Wilson 2012). State antitakeover laws afford managers legal protection against takeover threats and are documented to significantly shape managerial incentives in various corporate decisions such as financing (Garvey and Hanka 1999; Wald and Long 2007), investment and operation (Bertrand and Mullainathan 2003), voluntary disclosure (Zhao et al. 2013), and financial reporting (Armstrong et al. 2012). In this study, we examine whether the enhanced managerial control bestowed by state-imposed hurdles to hostile takeovers affects managers' incentives in tax management.

U.S. firms are governed by antitakeover statutes of their state of incorporation regardless of where they are headquartered and where they conduct business. Most U.S. states passed antitakeover legislation in the mid-to-late 1980s and early 1990s to shield firms incorporated in those states from takeover attacks. The state-level legal protection enhances managerial power and control, and has significant effects on managerial incentives and behavior. As an *externally* exerted governance control, antitakeover protection is documented by prior studies to have different impact in several aspects from *firm-level* internal governance. For example, Klein (2002) find that less effective board is associated with greater extent of earnings management while Armstrong et al. (2012) find that the enhanced managerial control after the passage of antitakeover laws reduces information asymmetry and improves firms' information environment. Karamanou and Vafeas (2005) find that firms with fewer outside directors disclose less while

Zhao et al. (2013) find that firms incorporated in states with more antitakeover laws have higher levels of voluntary disclosure.

Several studies investigate whether monitoring and discipline from firm-level governance mechanisms affect managerial incentive to pursue tax avoidance. Minnick and Noga (2010) report that CEO and director portfolio *delta* determines the level of tax avoidance, but other governance elements such as board composition, CEO duality, and firm-level antitakeover provisions are not influential. Rego and Wilson (2012) find that CEO option *vega* is related to risky tax management, while CEO option *delta* is not. Armstrong et al. (2015) find no evidence of the conditional mean or median of the relation between tax avoidance and board characteristics, but document a negative relation of financial sophistication and board independence with tax avoidance in the upper tail of the tax avoidance distribution, and a positive relation in the lower tail of the tax avoidance distribution. Overall, the extant evidence is mixed on the relation between firm-level governance control and the level of tax avoidance. Moreover, the evidence may not generalize to the discipline from the takeover markets.

To test the relation between state antitakeover protection and corporate tax avoidance, we use cash effective tax rates (ETRs) to proxy for the level of firm's tax avoidance, because cash ETRs capture the aggregate level of all types of tax avoidance, including both aggressive tax-avoidance activities and more conservative types of tax planning. In addition, cash ETRs are visible and understandable to investors. Taking account of year-to-year variations in annual effective tax rates, we measure cash ETRs over a one-year period and a three-year period. Following prior studies on the economic consequences of state antitakeover protection (e.g., Zhao et al. 2013; Mansi et al. 2009; Qi and Wald 2008; Wald and Long 2007; Francis et al.

2010), we use the number of antitakeover statutes in the state in which a firm is incorporated to measure the strength of legal protection against takeover.

Using a large sample of firms from 1987 to 2001, our univariate analysis indicates that firms incorporated in a state with strong antitakeover protection exhibit a lower level of tax avoidance in almost all the sample years. The negative relation between the strength of state antitakeover laws and the level of corporate tax avoidance is corroborated by a multivariate analysis with controls for firm-level attributes, year and industry fixed effects, state-of-location fixed effects, and time-varying industry and state-of-location effects. A quantile analysis reveals that the negative relation holds almost across the spectrum of tax avoidance. We continue to observe this relation after endogenizing firm decisions regarding where to incorporate using the Heckman (1979) two-stage procedure and the propensity score-matching approach. In addition, using firms as their own controls, we find that the level of tax avoidance decreases following the passage of state antitakeover statutes. Finally, our findings are not a manifestation of Delaware incorporation and are robust to the use of alternative effective tax rates and aggressive tax position measures. The strong negative relation suggests that enhanced managerial control accorded by antitakeover laws reduces managers' incentives to pursue tax avoidance.

We next explore the source of the negative relation by analyzing two plausible explanations. The simplest one is managerial slack: self-interested managers under alleviated takeover threat induced by state antitakeover protection may pursue a quiet life by avoiding costly efforts to maximize tax planning opportunities (Bertrand and Mullainathan 2003). However, our empirical evidence does not support the quiet life hypothesis, as the negative relation is not stronger in situations where managers are more likely to enjoy managerial slack (i.e., when firms operate in non-competitive industries or make fewer investments).

The second explanation lies in tax-related rent extraction. Tax avoidance may facilitate managerial rent extraction by providing managers with the means, masks, and justification to engage in opportunistic activities (Desai 2005; Desai and Dharmapala 2006). Increased entrenchment afforded by state antitakeover protection reinforces the shareholder perception of tax-related rent extraction, thereby increasing the likelihood of price discount by the market. Managers' concerns over stock price discount may prompt them to refrain from high levels of tax avoidance activities. We find evidence in support of the price discount concern hypothesis, as the negative relation is more pronounced when the concerns over stock price discount are heightened (i.e., when firms operate in industries with high external finance dependence). In the takeover market setting, one of top managers' concerns is their inability to re-contract away the adverse consequences of exacerbated agency problems when shielded by state antitakeover laws (Karpoff and Malatesta 1989;1995). The price-discount concern hypothesis is generally consistent with this takeover market phenomenon.¹

Prior studies examine the impact of firm-level antitakeover protection on firm's tax planning incentives and find little evidence that the Gompers et al. (2003) G-Index is significantly related to tax avoidance measures (Minnick and Noga 2010; Desai and Dharmapala 2006). The G-Index comprises 22 firm-level provisions and two state-level provisions, only one of which (business combination laws) is included in our measure of state antitakeover protection. Bebchuk et al. (2009) argue that the provisions in the G-Index may not be independently determined—some provisions might be the endogenous result of the allocation of power between shareholders and managers set by other provisions, and thus not all provisions bear the same relevance. Consistent

¹ Dual class ownership structure enhances insiders' control and insulates firms from hostile takeovers (Bebchuk and Cohen 2005). McGuire et al. (2014) fail to find that dual class firms undertake less tax avoidance for concerns over stock price discount, possibly because those firms opt for dual class ownership structure due to their more emphasis on control and less emphasis on stock prices (Smart and Zutter 2003).

with this view, we find that some provisions are positively, and some negatively, related to tax avoidance, and the others have little relevance. Giving the same weight to all provisions that load differently with tax avoidance obscures the effect of firm-level antitakeover protection on tax avoidance. As suggested by Bebchuk et al. (2009), it would be more practicable and useful in research to measure governance based on a few key provisions than an all-inclusive measure. We focus on a small number of state-level antitakeover provisions and find compelling evidence that state legal protection against takeovers affects firm's tax avoidance.

Our study makes three contributions to the finance and accounting literature. First, we add to the research on the effect of corporate governance on firm's tax planning decision. In contrast to prior studies that examine the impact of firm-level internal governance on tax avoidance, we focus on the effect of the takeover market, an *externally* exerted governance control, on managers' tax planning incentives and document a strong negative relation between antitakeover protection and tax avoidance. In addition, we explicitly address firms' endogenous incorporation decisions by employing both the Heckman (1979) two-stage procedure and the propensity score-matching approach. The takeover market setting also allows us to resort to the passage of state antitakeover statutes to implement a changes analysis to infer a causal relation between takeover protection and tax avoidance. More importantly, we explore the underlying mechanisms of the relationship. Given more than one explanation for the positive relation between the strength of governance control and tax avoidance, this attempt is important. Our study, together with prior ones, highlights that not all forms of governance mechanisms influence managerial tax planning decisions in the same manner.

Second, our study complements the literature that examines the economic consequences of state antitakeover laws on corporate decisions by adding firms' tax policies to

the affected list.² Lastly, we contribute to the tax literature by documenting that control considerations influence managers' incentives and preferences for tax planning activities and thus serve as an important determinant of tax avoidance.

The rest of the paper proceeds as follows. Section II discusses related literature and develops hypothesis. Section III details data, sample selection, and research design. Section IV offers primary empirical analyses and results on the relation between state antitakeover laws and corporate tax avoidance, and Section V provides explanations for the association. We conclude the paper in Section VI.

II. RELATED LITERATURE AND HYPOTHESIS DEVELOPMENT

Our main objective is to analyze the role of state-imposed hurdles to hostile takeovers in predicting the cross-section of firms' tax avoidance activities. In particular, we test two competing perspectives about the relation between the strength of state antitakeover defenses and the level of firm's tax avoidance. The investment opportunity perspective characterizes tax planning as one of the alternative risky investment opportunities that should be pursued to maximize firms' expected discounted after-tax cash flows. The rent extraction perspective views tax avoidance activities potentially serving as a means to enable and mask managerial rent extraction.

Investment Opportunity Perspective

The agency theory describes a natural conflict between shareholders and managers. The conflict arises when managers attempt to maximize their own utility function without fully pursuing all investment opportunities to maximize shareholder wealth. According to Jensen and

² Prior studies investigate the impact of state antitakeover laws on stock prices (Karpoff and Malatesta 1989, 1995), takeover probability (Comment and Schwert 1995), firm capital structure (Garvey and Hanka 1999), managerial ownership (Cheng et al. 2005), executive compensation (Cheng and Indjejikian 2009), firm operation and investment (Bertrand and Mullainathan 2003), corporate disclosure (Zhao et al. 2013), cost of debt financing (Francis et al. 2010), and information environment (Armstrong et al. 2012).

Meckling (1976), self-interested managers tend to enjoy managerial slack and avoid undertaking cognitively difficult tasks. Tax planning entails efforts and involves risks, which are costly to managers (Phillips 2003; Rego and Wilson 2012). Therefore, if not incentivized and/or pressured by shareholders, managers may attempt to avoid difficult decisions and costly efforts associated with tax avoidance. Consistent with this notion, Rego and Wilson (2012) find that economic incentives motivate CEOs to undertake risky tax strategies and Cheng et al. (2012) document that hedge fund activists press effort-averse managers to pursue tax planning opportunities.

There are other incentives and governance controls that can help prompt managers to pursue tax planning opportunities. For example, managers' fear of a hostile takeover may incentivize them to undertake value-maximizing tax planning activities. The legal protection accorded by state antitakeover statutes increases the difficulty of mounting a successful takeover and insulates managers from shareholder pressures, which allows them to pursue goals other than maximizing shareholder wealth (Comment and Schwert 1995). Bertrand and Mullainathan (2003) examine firms' operating and investing activities after the passage of second-generation antitakeover laws and find that firms incorporated in those states increase workers' wages, delay the closure of old plants and the start of new ones, and exhibit lower profitability and productivity. These are consistent with the "quiet life" preference, in which managers, once shielded from takeovers, are reluctant to perform cognitively difficult tasks. As the takeover protection increases managerial slack, it is expected that managers prefer to avoid difficult tasks related to tax planning. Accordingly, the quiet life hypothesis predicts a negative relation between the strength of state antitakeover protection and the extent of corporate tax avoidance.

Tax-related Rent Extraction Perspective

In contrast to the investment opportunity view that firms should pursue all tax avoidance opportunities, the rent extraction perspective recognizes the potential costs associated with tax avoidance and suggests that tax avoidance is value enhancing only in certain circumstances. Specifically, Desai (2005) and Desai and Dharmapala (2006) propose a complementary relation between tax avoidance and managerial rent extraction, in which strategies to minimize tax liabilities may advance interests of managers in ways that are detrimental to shareholders. Tax avoidance activities necessitate organizational and operational complexity and require opaque reporting to avoid being detected by tax authorities, which provides opportunities for managers to extract private benefits. Given this situation, tax avoidance in essence facilitates and masks rent extraction. Moreover, managers can justify their rent extraction on the pretense of minimizing firm tax burdens to serve shareholder interests. Hence, tax avoidance transactions may provide managers with the means, masks, and justifications to engage in opportunistic activities (Desai 2005; Desai and Dharmapala 2006).

Anecdotal evidence provides support to the notion of complementarity between tax avoidance and rent extraction. The cases of Tyco, Enron, and Dynegy, for example, show that active tax management strategies enhanced managers' ability to extract rents and obscure self-dealings. More importantly, empirical evidence shows that shareholders seem to understand tax-related rent extraction and associate tax avoidance with managerial opportunism. For example, Desai et al. (2007) find that an increase in tax enforcement in Russia receives favorable market reactions but decreases the voting premium of large shareholders, suggesting that investors view tax avoidance activities as controlling shareholders' expropriation of minority shareholders. Hanlon and Slemrod (2009) document a negative market response to the announcements of tax sheltering activities by poorly-governed firms. Desai and Dharmapala (2009) find a positive

relation between tax avoidance and firm value only for a subset of well-governed firms, which implies that investors perceive tax avoidance beneficial only when managerial rent extraction is minimal.

Because investors perceive firm's tax avoidance activities as being related to rent extraction, they are likely to price-protect themselves by discounting the stock price of a firm that is aggressive in tax planning (Chen et al. 2010). This likelihood grows in severe agency problem situations where shareholders have difficulty in distinguishing managerial motives of tax minimization from those of rent extraction. State antitakeover protection exacerbates the agency problems and prior studies find that shareholders are unable to re-contract away the adverse agency effects of antitakeover laws and that managers cannot creditably commit to avoiding rent extraction. Karpoff and Malatesta (1989, 1995), for example, find that the passage of state antitakeover laws on average is associated with decreases in stock prices of firms incorporated in the state. As such, managers' concerns over stock price discount may constrict their tax avoidance effort. Thus, the price discount concern hypothesis also predicts a negative relation between state antitakeover laws and tax avoidance.

Given that the two hypotheses predict a negative relation between state antitakeover laws and tax avoidance, we gauge their relative importance by conditioning the negative relation on different firm and industry attributes. If the quiet life hypothesis holds, the negative relation should be stronger when managers are more likely to enjoy managerial slack or actually pursuing the "so-called" quiet life; and if the price discount concern hypothesis is valid, the negative relation should be more pronounced when the concerns over stock price discount are heightened.

III. RESEARCH DESIGN

Sample Selection

Our sample period begins in 1987 because the second-generation antitakeover laws were ruled to be constitutional that year and the major tax reform (the Tax Reform Act of 1986) took effect in 1986. Our sample ends in 2001 that is the last year we have data on the state antitakeover laws index from Lucian Bebchuk's website.³ We start with all U.S. public companies in Compustat during the sample period and remove financial and utility firms as they are subject to different incentives due to regulation. We also require data availability on control variables, tax avoidance measures, and the state of incorporation. Our final sample consists of 22,945 firm-year observations of 5,615 distinctive firms.

The state of incorporation is essential for our analysis as it determines the antitakeover statutes to which firms are subject. Compustat industrial files are the main source of incorporation information; however, the dataset only reports firms' current state of incorporation. We obtain firm reincorporation information from Compustat CSTHIST header file and RiskMetrics Corporate Governance database (IRRC). Consistent with prior evidence that firm reincorporation has been rare (Bertrand and Mullainathan 2003), we find that only 143 of our sample firms underwent changes in their state of incorporation. We use historical incorporation information for these firms.⁴

Measure of State Antitakeover Protection

As with prior studies on the economic consequences of state antitakeover laws (Zhao et al. 2013; Mansi et al. 2009; Qi and Wald 2008; Wald and Long 2007; Francis et al. 2010), we use the number of antitakeover statutes in the state in which a firm is incorporated to measure the

³ See <http://www.law.harvard.edu/faculty/bebchuk/data.shtml>.

⁴ We obtain similar results if we follow Zhao et al. (2013) and eliminate firm-years for firms with changes in the state of incorporation.

strength of state-level antitakeover protection.⁵ We follow Zhao et al. (2013) and redefine the Bebchuk and Cohen (2003) five-provision index by considering two additional, more restrictive statutes—the recapture statutes adopted by Ohio and Pennsylvania and the staggered board statute by Massachusetts.⁶ We further adjust the index value for each firm-year based on a firm’s opt-out decision in that year as this decision helps recover shareholder wealth losses associated with the passage of antitakeover statutes (Szewczyk and Tsetsekos 1992).

Table 1 reports the average value of antitakeover index by state, the number of unique firms incorporated in each state, and corresponding firm-years. One notable observation from Table 1 is the large variations in the number of antitakeover statutes across different states. This should serve us well to detect the effect of antitakeover protection on tax avoidance. Consistent with prior studies, more than half of our sample firms are incorporated in the state of Delaware. We test whether our results are driven by these firms in the later analysis.

Tax Avoidance Measures

We are interested in inquiring into how the strength of state-level antitakeover protection affects firms’ incentives to undertake tax avoidance activities. Hence, the appropriate proxies for our research question should be the ones that capture a wide variety of tax management strategies from benign and legal tax avoidance to aggressive tax planning practices. Both GAAP effective tax rates (ETRs) and cash effective tax rates capture the aggregate level of tax

⁵ On one hand, more antitakeover provisions represent stronger deterrents to hostile takeovers and greater protection for incumbent management. On the other hand, the adoption of several types of statutes might have conveyed a message that the state’s corporate law is committed to providing substantial protections from takeovers (Bebchuk and Cohen 2003).

⁶ The five standard types of antitakeover provisions, taken from Gartman (2000), include control share acquisition statute, fair price statute, business combination statute, poison pill endorsement statute, and constituencies statute. A control share statute requires a hostile bidder to put its offer to a vote of shareholders early in the process. A fair-price statute requires a bidder who gains control to pay remaining minority shareholders the same price it paid for shares acquired in the original bid. Business combination statutes restrict bidders on merging assets for a specified number of years. Poison pills are rights that entitle existing holder to get significant value in the event of an acquisition without board approval. Constituencies statutes allow managers to take into account interests of non-shareholders in defending against a takeover. Our results remain unchanged if we use Bebchuk and Cohen (2003) five-provision index.

avoidance and are measures visible to investors and regulators as evidence of high levels of tax avoidance (McGuire et al. 2014). However, GAAP ETRs mostly reflect managers' objective to reduce income tax expense for financial reporting purposes whereas cash-based ETRs reflect managers' effort to reduce actual cash tax burden (Dyreng et al. 2010). Accordingly, we use cash-based ETRs as our primary tax avoidance measure.

Cash ETRs have several advantages in addressing our research question. First, cash ETRs capture both permanent and temporary tax avoidance activities while GAAP ETRs exclude the tax effects of temporary deferral strategies (Dyreng et al. 2008). Second, unlike GAAP ETRs which may be biased by changes in book accruals within tax expense such as valuation allowance for unlikely realized deferred tax assets and tax contingency for unsustained tax positions, cash ETRs are not affected by changes in tax accounting. Third, cash ETRs avoid overstatement of current tax expense due to accounting for the income tax benefits of employee stock options during our pre-SFAS 123R sample period (Hanlon and Shevlin 2003).

Since isolated events may impact one-year cash ETRs, we measure cash ETRs over a three-year period as well. Following Dyreng et al. (2008), we estimate one-year cash ETR (*CashETR*) as the firm's cash taxes paid in current year divided by pre-tax income adjusted for special items in the same year, and three-year cash ETR (*CashETR3*) as the total cash tax paid over the three years ending in year *t* divided by pre-tax book income excluding special items summed over the same three-year period. Large (small) values of *CashETR* and *CashETR3* correspond to lower (higher) levels of tax avoidance. Consistent with prior literature, we require our observations to have positive pre-tax book income before special items. We further winsorize ETR values at zero and one.

Baseline Regression Model

To isolate the effect of state antitakeover protection on corporate tax avoidance, we model tax avoidance as a function of state antitakeover index and firm-level factors identified by prior studies to be related to tax avoidance. The function is specified as follows:

$$\begin{aligned} TaxAvoidance_t = & \beta_0 + \beta_1 StateLawIndex_t + \beta_2 SizeAt_t + \beta_3 Accrual_t + \beta_4 ROA_t + \beta_5 NOL_t \\ & + \beta_6 ChgNOL_t + \beta_7 Leverage_t + \beta_8 Cash_t + \beta_9 FIncome_t + \beta_{10} R\&D_t \\ & + \beta_{11} PPE_t + \beta_{12} Intangible_t + \beta_{13} EQING_t + Year-fixed\ Effects \\ & + Industry-fixed\ Effects + State-of-location-fixed\ effects + \varepsilon_t, \end{aligned} \quad (1)$$

where the dependent variable, *TaxAvoidance*, represents one of the two tax avoidance measures (*CashETR* and *CashETR3*). The variable of interest is *StateLawIndex*, and we expect a positive coefficient on *StateLawIndex* to support our hypotheses.

Guided by prior studies, our choices of control variables aim at controlling for the effects of firm size, accounting quality, firm performance, foreign operations, and other firm characteristics (e.g., Mills et al. 1998; Rego and Wilson 2012; Dyreng et al. 2010, 2008; Chen et al. 2010). Detailed definitions of these control variables are presented in Appendix and all continuous variables are winsorized at the 1th to 99th percentiles. We include the natural logarithm of total assets (*SizeAT*) to control for firm size as, on one hand, large firms may be reluctant to avoid taxes due to political cost concerns (Zimmerman 1983), and on the other hand, large firms are more sophisticated in structuring complex tax-reduction transactions (Mills et al. 1998; Hanlon et al. 2007). Prior studies find that multinationals place income in lower taxed jurisdictions (Klassen and Laplante 2012) and/or place debt in high-tax locations (Newberry and Dhaliwal 2001) to reduce tax burdens. We control for the effect of foreign operations by including the ratio of foreign pretax income to total assets (*FIncome*). The inclusion of cash ratio (*Cash*) is to control for liquidity as Edwards et al. (2013) find that firms' incentives to undertake tax saving initiatives are affected by their financial constraints.

Tax avoidance and earnings quality are interrelated as Frank et al. (2009) find that firms aggressive in financial reporting are tax aggressive as well. Firms also respond to state-level antitakeover protection by voluntarily improving financial reporting quality (Armstrong et al. 2012). To ensure that the association between antitakeover protection and tax avoidance is not driven by earnings quality, we include in the regression the absolute value of discretionary accruals estimated from Modified Jones Model. Similarly, since firm leverage is negatively related to both effective tax rates (e.g., Mills and Newberry 2004) and increased legal barriers to takeover (Garvey and Hanka 1999), we include leverage ratio in the model to ensure that the impact of antitakeover statutes on tax avoidance is incremental to the effect of leverage.

The inclusion of pretax return on assets (*ROA*) and loss carryforwards (*NOL* and *ChgNOL*) addresses the effects of profitability and utilization of loss carryforwards on cash taxes paid (Chen et al. 2010). Given that intellectual property increases firms' opportunity to reduce taxes via income shifting and R&D tax credit reduces cash ETRs (Edwards et al. 2013), we include research and development expense (*R&D*) as a control variable. As different treatments of depreciation expenses for financial and tax reporting purposes affect cash ETRs and the effect is particularly pronounced for capital intensive firms, we include the ratio of net property, plant and equipment to total assets (*PPE*) to control for capital intensity. Similarly, we include intangible assets (*Intangible*) and equity in earnings (*EQINC*) in the model to control for differential book and tax treatments of intangible assets and consolidated earnings accounted for using the equity method (Chen et al. 2010).

Because cash ETRs may vary across industries (Dyreng et al. 2008), we control for industry-fixed effects with industry dummies based on the Fama and French (1997) 48 industry classification. Following prior studies, we control for differences across the states in which the

sample firms are headquartered by adding a set of state-of-location dummy variables. As firms respond to economic shocks by increasing internally generated funds via tax planning (Edwards et al. 2013), we control for macroeconomic factors that affect tax avoidance incentives for all our sample firms in the year by including year-fixed effects. All continuous variables are winsorized at the 1 and 99th percentile and all standard errors adjusted for clustering at the firm-year level to account for serial correlations across years within the same firm.

IV. EMPIRICAL RESULTS

Descriptive Statistics

Panel A of Table 2 contains descriptive information of tax avoidance, state antitakeover index, and control variables used in our analyses. The mean (median) value of one-year cash ETR is 0.299 (0.286) and of three-year cash ETR 0.302 (0.301). These statistics are within the range reported by prior studies.⁷ Regarding the measure of antitakeover protection, the mean (median) antitakeover index is 2.069 (1.000), suggesting that on average a sample firm is protected by two state antitakeover statutes. The distribution of the index is compatible to that reported by Zhao et al. (2013) even though our sample is much larger. With respect to control variables, the logarithm of total assets averages 5.045, the absolute discretionary accrual 0.071, pretax rate of return 0.124, the ratio of long-term debt to total assets 0.214, cash ratio 0.158, the ratio of pretax foreign income to total assets 0.008, and the ratio of equity income in earnings to total assets 0.001. About 20 percent of the firms report loss carryforwards. R&D expense, PPE assets and intangible assets account for 3.0 percent, 35.9 percent and 11.4 percent of total assets, respectively. These statistics are consistent with prior studies (e.g., McGuire et al. 2014).

⁷ For example, the mean (median) one-year cash ETR is 0.270 (0.256) in Dyreng et al. (2010), 0.27 (0.28) in Rego and Wilson (2012), 0.275 (0.240) in Koester et al. (2013), and 0.306 (0.301) in McGuire et al. (2014). Our measure is a little higher than that reported by prior studies mainly because our sample starts earlier and cash ETR is higher in earlier year. If we estimate the measure over 1994–2001, the mean (median) annual cash ETR is 0.287 (0.275).

Panel B of Table 2 reports Pearson pairwise correlations among variables used in this study. Three observations emerge from the correlation table. First, one-year cash ETR (*CashETR*) is highly correlated with three-year cash ETR (*CashETR3*). Second, state antitakeover index is positively correlated with both measures of cash ETR, providing preliminary evidence on the negative relation between state antitakeover protection and tax avoidance. Third, correlations between explanatory variables are generally small (all correlation coefficients are below 0.50), suggesting little concern for multicollinearity problem.

Univariate Analysis

We first take a look at the data by performing univariate comparisons of the two cash-based ETRs between firms with strong antitakeover defenses and those with weak antitakeover protection on a yearly basis. A firm falls into the weak subgroup if incorporated in a state with antitakeover index less than 3.0 and into the strong subgroup otherwise. As shown in Table 3, one-year cash ETR, on average, is 0.319 for the strong subgroup and 0.289 for the weak subgroup, and the difference is significant at the 1% level. Similarly, the difference in the average value of three-year cash ETR is significant as well (0.317 vs. 0.294, $t = 8.58$).

Year-by-year comparisons show that in all 15 years of the sample period the strong subgroup reports higher annual cash ETRs than the weak subgroup, and the differences are statistically significant in 11 years. The findings for three-year cash ETR are similar. Figure 1 plots annual averages of cash ETRs for strong/weak subgroups over our sample period. Both Table 3 and Figure 1 indicate that firms incorporated in states with strong antitakeover defenses pay more taxes relative to their pretax income than their counterparts incorporated elsewhere and that this difference is not driven by time clustering effects. Overall, the univariate results suggest

that firms incorporated in states strong in antitakeover protection exhibit a lower tendency to avoid taxes than firms incorporated in states with weak antitakeover protection.

Baseline Regression

Since the univariate analysis suffers from omitted variable bias, we examine the relation in a multivariate setting by estimating Equation (1) using cross-sectional time-series data. Estimated results reported in Panel A of Table 4 show that the coefficient on *StateLawIndex* is positive and significant at conventional levels for both cash-based ETR measures, suggesting an incremental effect of state antitakeover protection on managers' incentives to undertake tax avoidance activities beyond that of other factors identified by prior studies. The coefficient of 0.005 on *StateLawIndex* when the dependent variable is *CashETR* implies that one-year cash ETR for firms incorporated in states with six antitakeover statutes is, on average, 0.030 higher than that for firms incorporated in nonrestrictive states (no antitakeover statute) and the comparable number for three-year cash ETR is 0.018. Considering the mean value of 0.299 (0.302) for *CashETR* (*CashETR3*), the differences are nontrivial.⁸ These corroborate the univariate findings that firms under strong state antitakeover protection tend to pursue less tax avoidance and thus report higher cash effective tax rates.

As to the coefficients on control variables, we find that large firms engage in lower levels of tax avoidance, in support of the political cost concern argument. Consistent with loss carryforwards reducing taxable income and thus tax burdens, we document a negative relation between loss carryforwards and cash ETRs. The negative coefficient on leverage supports previous evidence that use of debts reduces effective tax rates. Finally, we report a negative relation of cash ETRs with R&D expense and long-term fixed assets, which supports the idea

⁸ Chen et al. (2010) find that family firms report 0.012 higher cash effective tax rate than non-family firms.

that firms with intellectual property and capital-intensive firms have more opportunities to engage in tax avoidance.

Quantile Regression

The OLS estimates reflect the effect of state antitakeover protection on the *average* level of tax avoidance. To shed more light on the nature of relation between the takeover market and tax avoidance, we look into finer results by investigating whether the impact of state antitakeover laws on tax avoidance varies at different points of the tax avoidance distribution. Specifically, we follow Armstrong et al. (2015) and estimate Equation (1) using quantile regression. The coefficient estimates in Panel B of Table 4 (coefficients on control variables are not tabulated for brevity) show that the relation between state antitakeover defenses and tax avoidance is negative and significant in most points of the tax avoidance distribution, suggesting a strong negative impact of state antitakeover laws on firms' tax avoidance. As a comparison, Armstrong et al. (2015) find that CEO option *vega* has a strong positive impact on tax avoidance across most quantiles, but the impact of board characteristics concentrates at the two extreme tails.

Another observation from the quantile regression analysis is that the magnitude of negative impact of state antitakeover protection decreases in the descending distribution of tax avoidance (ascending distribution of cash ETRs), and the impact turns positive at the lowest level of tax avoidance, suggesting that aggressive tax planning drives the negative relation between state antitakeover laws and tax avoidance. Because managers pursuing the so-called "quiet life" likely underinvest in tax avoidance, that is, quiet-life seekers tend to locate in the lower range of tax avoidance, the quiet life hypothesis predicts a stronger negative relation between antitakeover protection and tax avoidance in the lower tail of the tax avoidance distribution than in the upper range. However, our results do not appear to be consistent with the quiet life hypothesis. Rather,

they seem to fit well into the price discount concern hypothesis. State antitakeover protection has stronger impact on tax avoidance at the upper tail of its distribution because of greater concerns over stock price discount as investors perceive increased likelihood of rent extraction when firms take risky tax positions. By contrast, price discount concerns are minimal when investors perceive rent extraction as less likely at the extreme lower tail of the tax avoidance distribution.

Endogeneity of Incorporation Decisions

Certain states reportedly passed antitakeover laws in response to promotions by large firms incorporated in those states;⁹ Romano (1987), however, argued that the adoption of antitakeover statutes typically did not result from pressures of a large coalition of economic players in the states. This exclusive nature of the lobbying process should help alleviate some concerns about the endogeneity of the state's antitakeover laws. However, as U.S. firms can choose a state other than their home state to incorporate based on their characteristics and antitakeover protection strength of their home and incorporation states (Bebchuk and Cohen 2003; Wald and Long 2007), incorporation decisions may be endogenous to tax avoidance—tax avoidance incentives may drive firms' incorporation decisions. In addition, to the extent that both firms' tax avoidance incentives and incorporation decisions are affected by unobservable factors, those factors may drive the negative relation between state antitakeover protection and tax avoidance even if state antitakeover protection *per se* has no impact on tax avoidance. In this section, we employ three methods to address the endogeneity issue. We first use the Heckman (1979) two-stage procedure to correct for potential self-selection bias arising from unobservable factors, then take the propensity score-matching approach to tackle endogenous incorporation

⁹ For example, the business combination law in Connecticut was promoted by Aetna Life and Casualty Insurance Company, a corporation incorporated in the state. The Arizona business combination statute was originally written by Greyhound executives.

decisions arising from observable factors, and finally conduct a changes analysis by examining changes in tax avoidance following the passage of antitakeover laws.

To implement the Heckman (1979) two-stage and propensity score-matching estimation effectively, we need at least one instrumental variable that is correlated with incorporation decisions but uncorrelated with tax avoidance decisions. We use firms' number of employees and home state population as instruments. As argued and indicated by Wald and Long (2007) and Zhao et al. (2013), firms with more employees are more likely to incorporate in their home states because of clout. As firms incorporated in small states may choose to locate in large states with more population (e.g., to obtain market and skillful labor), state population affects in-state versus out-of-state incorporation decisions—home state and incorporation state are more likely to differ for firms located in large state. There is no evidence indicating that firms' number of employees and home state population affect tax avoidance.

We start with the first-stage regression in which we model firms' decisions to incorporate in-state versus out-of-state as a function of firm size, profitability, R&D expenditure, advertising expense, capital intensity, depreciation expense, home state antitakeover index, home state total asset constraint for payout, number of employees, home state population, and industry and year fixed effects. We estimate a Probit regression with *HomeState* (an indicator variable equal to one if a firm is incorporated in its home state, and zero otherwise) as the dependent variable. Similar to prior studies (Zhao et al. 2013), we find that roughly one-third of our sample firms incorporate in their home states (7,775 among 21,916 firm-years). The estimated results from the first-stage regression are presented in Panel A of Table 5. Consistent with home state population and firms' number of employees affecting incorporation decisions, we document a negative coefficient on state population and a positive coefficient on number of employees. We also find that large firms

in terms of total assets are more likely to incorporate out of state, indicating that number of employees and total assets capture different dimensions of firm scale. The results further show that firms are more likely to incorporate in home states if antitakeover protection in home states is strong. The pseudo R^2 of 53.74 percent indicates a high explanatory power of our model for incorporation decisions.

For the Heckman (1979) two-stage estimation, we calculate the inverse Mills ratio (*InvMills*) from the first-stage selection regression and include it in the second stage outcome model to correct for selection bias. Panel B of Table 5 reports the second-stage estimates. The coefficient on *StateLawIndex* remains positive and significant for both tax avoidance measures, and the coefficients on control variables remain almost unchanged. Markedly insignificant is the coefficient on *InvMills*, which indicates that incorporation decisions are not endogenous to tax considerations.

To implement the propensity score-matching estimation, we find an out-of-state incorporation control firm for each in-state incorporation firm based on the predicted probabilities from the first-stage regression. The search algorithm finds the propensity scores closest to the in-state incorporation firm observation in the same year until the first decimal place. The control firm constructed in this way has the same predicted probabilities of incorporating in home state but in fact incorporates out of home state. The propensity score-matching produces a matched sample of 2,260 control firm observations, resulting in a combined sample of 4,520 firm-years. We conduct a covariate balance test to assess the effectiveness of matching. As shown in Panel C of Table 5, the treatment sample (in-state incorporation firms) is similar to the matched control sample (out-of-state incorporation firms) in almost all dimensions with the exception of R&D expenditure.

We apply the matched sample to Equation (1) and report the coefficient estimates in Panel D of Table 5. The coefficients on *StateLawIndex* are still positive and significant, and their magnitudes even greater than those estimated from the OLS regression as shown in Table 4. The overall results suggest that the negative relation between tax avoidance and state antitakeover laws is not driven by endogenous incorporation decisions.

Using firms as their own controls, we examine changes in the extent of tax avoidance following the passage of state antitakeover statutes, an exogenous shock to the takeover market. Following prior studies (Bertrand and Mullainathan 2003; Armstrong et al. 2012) we create an indicator variable, *PassageBCLaw*, based on the passage of the business combination laws. It takes the value of one for years following the legislation and zero otherwise. We estimate Equation (1) by replacing *StateLawIndex* with *PassageBCLaw* for the sample firms that exist in both pre- and post-passage periods. As in Armstrong et al. (2012), we control for the time-varying industry and state-of-location effects by including industry-year- and state-year-adjusted tax avoidance rather than industry and state fixed effects to separate out the effects of industry and geographic shocks that are contemporaneous with the antitakeover laws from the effects of the laws themselves. The estimated results are reported in Panel E of Table 5. The coefficient on *PassageBCLaw* is 0.014 (p-value = 0.056) for *CashETR* and 0.022 (p-value = 0.001) for *CashETR3*, suggesting that an exogenous increase in state antitakeover protection decreases the extent of corporate tax avoidance.¹⁰

Excluding Firms Incorporated in Delaware

Since most of our sample firms are incorporated in Delaware, one concern is that our results may only reflect the effect of incorporation in Delaware rather than the impact of

¹⁰ For the post-passage period, we exclude observations for the first two years following the passage of the BC law to avoid overlapping with the pre-passage period when the dependent variable is *CashETR3*.

antitakeover laws. To abstract from the Delaware effect, we drop the Delaware firms from the sample and re-run Equation (1). The estimated results, reported in Panel A of Table 6, indicate that the impact of antitakeover laws on tax avoidance exists in other states. Specifically, the coefficient on *StateLawIndex* is 0.006 (p-value = 0.006) and 0.006 (p-value = 0.001) for *CashETR* and *CashETR3*, respectively, for this subsample.

Alternative Tax Avoidance Measures

As stated earlier, cash-based ETRs are the most appropriate measures for our research question. However, the statistics in Panel B of Table 6 show that the documented negative relation between state antitakeover laws and tax avoidance remains robust to alternative effective tax rates such as GAPP effective tax rates and current effective tax rates from Porcano (1986).¹¹ We also assess whether our results are robust to risky tax positions proxied by total book-tax differences of Frank et al. (2009) and estimated tax sheltering probability of Lisowsky (2010).¹² We find that the strength of antitakeover laws is negatively associated with risky tax positions.

Firm-level Antitakeover Protection

Prior studies find little evidence that firm-level antitakeover protection proxied by the G-Index (Gompers et al. 2003) affects the extent of tax avoidance (Minnick and Noga 2010; Desai and Dharmapala 2006). The G-Index comprises 22 firm-level provisions and two state-level provisions (business combination laws and control-share cash-out laws). The irrelevance of the G-Index to tax avoidance may be because not all provisions in the index load in the same way with tax avoidance measures. To test whether this is the case, we estimate Equation (1) with each

¹¹ The GAAP ETR is estimated as income tax expenses divided by pretax income, and current effective tax rate from Porcano (1986) is estimated as current federal tax expense / (income before extraordinary items + current federal tax expense + minority interest – extraordinary items – equity in earnings).

¹² Frank et al. (2009) total book-tax differences is calculated as (pre-tax income – (current federal tax expense + current foreign tax expense) / Statutory tax rate) / lagged assets) and the Lisowsky's (2010) estimated tax sheltering probability is computed as the predicted value of the model in Column 2 of Table 4 in Lisowsky (2010).

firm-level provision as the variable of interest (replacing *StateLawIndex*) and find that blank check preferred stock, compensation plans, golden parachutes, and poison pills are negatively, and limited ability to amend bylaws, supermajority to approve merger, and directors' duties positively, related to cash ETRs, and the remaining 15 irrelevant (five have negative signs and 10 positive signs, but none of them is significant). The tests validate the conjecture that not all provisions bear the same relevance to tax avoidance and the G-Index that assigns the same weight to each provision obscures the effect of antitakeover defenses on tax avoidance.

Given that the influence of state-level antitakeover protection on corporate decisions may depend on preexisting firm-level takeover defenses (Karpoff and Malatesta 1989; Garvey and Hanka 1999), we investigate whether state antitakeover laws have incremental effect on tax avoidance. Specifically, we control for the effects of dual class ownership and firm-level antitakeover provisions. The information of dual class ownership (1994-2001) is from the data set used in Gompers et al. (2003);¹³ and we measure antitakeover protection at the firm level using different combinations of the provisions in the G-Index, such as the G-Index itself, the Bebchuk et al. (2009) E-Index (based on six provisions), an index including provisions having negative signs with cash ETRs, and an index consisting of provisions having positive signs with cash ETRs. The results (untabulated) show that firms with dual class ownership exhibit lower levels of tax avoidance, consistent with McGuire et al. (2014), and that the G-Index and the E-Index are not significantly related to tax avoidance. More importantly, the coefficient on *StateLawIndex* remains positive and significant, suggesting that state-level antitakeover statutes have a real bite on firms' tax avoidance beyond firm-level antitakeover defenses.

Miscellaneous Robustness Checks

¹³ We thank Andrew Metrick for sharing the data set.

We assess the robustness of our documented results to the variants of antitakeover law index. To avoid imposing linearity on the effects of any given increase in the index score, we use an indicator variable to denote the strength of state antitakeover protection (equal to one if antitakeover index is not less than 3.0 and zero otherwise), and the results stay unchanged. We also note that the negative relation between state antitakeover laws and tax avoidance remain the same if we use the Bebchuk and Mullainathan (2003) five-measure index (without including recapture law) to proxy for antitakeover protection.

In our main regressions, we include industry and state-of-location fixed effects to control for time invariant industry and home state factors. If we control for time-varying industry and state-of-location effects by including industry-year- and state-year-adjusted tax avoidance, we obtain similar results, suggesting that our results are robust to the controls for effects of industry and geographic shocks. To further address the problem of clusters of observations within cross-sections, we replicate the main analysis using the Fama and MacBeth (1973) procedure by estimating Equation (1) annually (without year fixed effects) and take the average of annual coefficients. Our untabulated results remain qualitatively unchanged. To further mitigate the concern of outlier effects, we conduct a rank regression by ranking our continuous variables into percentiles each year and scale them to values between zero and one. Using these ranked variables to estimate Equation (1), we also obtain similar results.

V. UNDERLYING CHANNELS

Quiet Life Hypothesis

To test the quiet life effect, we identify situations in which managers are more likely to enjoy managerial slack or firms are actually pursuing the so-called "quiet life." Economists have long argued that product market competition suppresses managerial slack, and Giroud and

Mueller (2010) provide empirical evidence that managers in firms from non-competitive industries tend to enjoy the quiet life. Armstrong et al. (2012) treat managers as pursuing the quiet life if their firms make fewer investments and pursue fewer growth options, manifested in lower than industry and state average growth in investment. If the negative relation is explained by the quiet life hypothesis, this relation should be more pronounced for firms in non-competitive industries and firms making fewer investments.

We measure industry competition using the Herfindahl-Hirschman Index (HHI), computed as the sum of squares of the market share of each firm in an industry each year. Industry is defined in accordance with the Fama and French (1997) 48 industry classification and market share is computed based on sales. We create an indicator variable, *HHIndex*, based on the annual median values of HHI, with the value of one representing non-competitive industries and the value of zero denoting competitive industries. We follow Armstrong et al. (2012) and treat managers in firm-years with negative industry- and state-adjusted growth in total assets, sales, and property, plant and equipment as pursuing the quiet life. More than 37 percent of firm-years in our sample fall into this category. We use an indicator variable, *Quiet*, to denote the quiet life seekers.

We augment Equation (1) by adding *HHIndex* or *Quiet* and the interaction term of *HHIndex* or *Quiet* with *StateLawIndex*. If the quiet life hypothesis explains the negative relation, we would observe a positive and significant coefficient on the interaction term. As shown in Table 7, the coefficients on the interaction terms are not statistically significant for any tax avoidance and the quiet life measures. The results suggest no difference in the effect of state antitakeover protection on tax avoidance between firms that operate in competitive industries and those in non-competitive industries, and between firms that make fewer investments and those

that invest more. These findings further confirm that the quiet life hypothesis does not explain the negative relation between state antitakeover laws and tax avoidance.

Price Discount Concern Hypothesis

We test the relevance of the price discount concern hypothesis by investigating whether the negative relation is stronger when concerns over stock price discount are heightened. State antitakeover protection aggravates managerial entrenchment and enhances investors' perception of tax-related rent extraction. Worried about possible stock price discount by investors, managers may refrain from engaging in tax avoidance. The concerns should be particularly relevant for firms operating in industries with high dependence on external financing. Thus, the negative relation between tax avoidance and antitakeover laws should be more pronounced for firms in those industries.

We follow Rajan and Zingales (1998) to define external finance dependence. We first estimate external finance dependence at the firm level as capital expenditure minus funds from operations divided by capital expenditure. To construct an industry-level measure, we use the industry median at the three-digit SIC code level. Then we classify industries as high and low external finance dependence based on annual median values. *FinDep* is an indicator variable equal to one if a firm is in an industry with strong external finance dependence and zero otherwise. We add to Equation (1) *FinDep* and its interaction term with *StateLawIndex*. If the price discount concern hypothesis explains our findings, the coefficient on the interaction term *FinDep* * *StateLawIndex* should be positive.

As reported in Table 8, the coefficient on *StateLawIndex_t* * *FinDep* is 0.003 with a p-value of 0.048 when tax avoidance measure is *CashETR*, and 0.005 with a p-value of 0.001 when tax avoidance measure is *CashETR3*. The results are consistent with the notion that managers

voluntarily constrain tax avoidance activities when protected by strong state antitakeover laws, due to concerns over possible market penalties for perceived tax-related rent extraction. Their willingness to abate tax avoidance is stronger when the concerns over stock price discount are aggravated by greater external finance dependence, in support of the price discount concern explanation. Interestingly, the effect of antitakeover laws on tax avoidance also exists for firms with less concerns over tax-related price discount as the coefficient on *StateLawIndex* remains positive for one-year cash ETR, suggesting that other effects are also in play to influence the relation between state antitakeover laws and tax avoidance.

VI. CONCLUSIONS

State antitakeover laws significantly inhibit the takeover market and strengthen managerial power and control. In this study, we examine whether the enhanced managerial control affects their incentives to pursue tax avoidance. We identify two possible channels through which state antitakeover statutes can influence firms' tax planning incentives. The quiet life hypothesis argues that managers insulated from takeover threat tend to shy away from the difficulties associated with tax avoidance and thus underinvest in tax planning opportunities. The price discount concern hypothesis contends that increased entrenchment stemming from state antitakeover defenses hardens investors' perceptions of tax-related rent extraction and managerial concerns over stock price discount restrain from tax avoidance activities.

Using a large sample of firms from 1987 to 2001, we document a negative relation between antitakeover defenses of a state and the extent of tax avoidance of firms incorporated in the state. Further analyses indicate that the negative association does not stem from managers protected by antitakeover laws enjoying the quiet life by underinvesting in tax planning opportunities. Rather, our findings are consistent with the price discount concern hypothesis—

mangers refrain from tax planning due to concerns over enhanced investors' perception of tax-related rent extraction when state antitakeover laws exacerbate the traditional agency problems. Our results are consistent with the takeover market phenomenon that a top managerial concern is their inability to re-contract away the adverse consequences of exacerbated agency problems when shielded by state antitakeover laws.

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Appendix Variable Definitions

Measures of Tax Avoidance

CashETR _t	=	Dyreng et al. (2008) cash effective tax rate calculated as Cash taxes paid/Pretax income = TXPD/(PI-SPI).
CashETR3 _t	=	The three-year sum of cash taxes paid (TXPD) ending in year t dividing by the three-year sum of pretax income(PI – SPI) ending in year t.
GAAPETR _t	=	GAAP effective tax rate estimated as income tax expenses (TXT) divided by pre-tax income (PI).
CurCashETR _t	=	Current effective tax rate from Porcano (1986) estimated as current federal tax expense / (income before extraordinary items + current federal tax expense + minority interest – extraordinary items – equity in earnings) = TXFED / (IB + TXFED + MII – XIDO – ESUB).
TBTD _t	=	Frank et al. (2009) total book-tax difference calculated as (pre-tax income – (current federal tax expense + current foreign tax expense)/Statutory tax rate)/lagged assets = (PI - (TXFED+TXFO)/Statutory tax rate)/AT _{t-1} . If current federal tax expense (TXFED) is missing, then we calculate it as total tax expense - deferred tax expense - state tax expense, - foreign tax expense = TXT – TXDI – TXS - TXFO. If information for TXFO or TXS is missing, we set the value of each of them to zero.
ProTaxShelter _t	=	Lisowsky's (2010) estimated tax sheltering probability computed as the predicted value of the following regression model (Column 2 of Table 4 in Lisowsky (2010)): $\text{ProTaxShelter}_t = -43.47 + 0.756 \times \text{BTD}_t + 0.748 \times \text{DAccrual}_t - 1.036 \times \text{Leverage}_t + 0.70 \times \text{SizeAT}_t + 3.196 \times \text{ROA}_t + 17.305 \times \text{FIncome}_t + 4.391 \times \text{RDA}_t$ Where: BTD _t = book - tax difference defined; DAccrual _t = discretionary accruals from the performance-adjusted modified cross-sectional Jones Model; Leverage _t = firm leverage defined as long-term debt (DLTT _t) divided by total assets (AT _{t-1}); SizeAT _t = the log of total assets in year t; ROA _t = return on assets (PI _t /AT _{t-1}); FINCOME _t = foreign pre-tax income (PIFO _t) divided by total assets (AT _{t-1}); RDA _t = research and development expense (XRD) divided by total assets (AT _{t-1}).

Measures of State Antitakeover Statutes

StateLawIndex _t	=	The antitakeover index in year t measured as the number of state antitakeover statutes (control share statute, fair price statute, business combination statute, poison pill endorsement statute, constituencies statute, and recapture statute) in the state where the firm is incorporated.
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Control Variables for Baseline Regression

SizeAT _t	=	The Natural logarithm of total assets (AT _t).
Accrual _t	=	Absolute value of discretionary accruals, where discretionary accruals are computed using the modified Jones model including lagged ROA as an additional regressor.
ROA _t	=	Return on assets estimated as operating income divided by lagged total assets = (PI _t – XI _t) / AT _{t-1} .
NOL _t	=	An indicator variable coded as one if loss carry forward (TLCF) is positive, and zero otherwise.
ΔNOL _t	=	Yearly change in loss carry forward (TLCF _t – TLCF _{t-1}) / AT _{t-1} .
Leverage _t	=	Leverage ratio calculated as long-term debt (DLTT _t) divided by lagged total assets (AT _{t-1}).
Cash _t	=	Cash ratio calculated as cash and cash equivalent (CHE _t) divided by lagged total assets (AT _{t-1}).
FIncome _t	=	Foreign operating income (PIFO _t) divided by lagged total assets (AT _{t-1}). Missing values for PIFO are set to zero.
R&D _t	=	R&D expenses (XRD _t) divided by lagged assets (AT _{t-1}). Missing values for XRD are set to zero.
PPE _t	=	Net property, plant and equipment (PPENT _t) divided by lagged total assets (AT _{t-1}).
Tangible _t	=	Intangible assets (INTAN _t) divided by lagged total assets (AT _{t-1}).

EQINC _t	=	Equity income in earnings (ESUB _t) divided by lagged total assets (AT _{t-1}).
<u>Other Variables</u>		
Population _t	=	A firm's home state population in year t-1.
Employee/Population _t	=	A firm's number of employees (EMP) divided by home state population.
CashFlow _{t-1}	=	Operating income before depreciation (OIBDP _{t-1}) divided by total assets (AT _{t-1}).
Advertise _{t-1}	=	Advertise expense (XRD _{t-1}) divided by total assets (AT _{t-1}).
Depreciation _{t-1}	=	Depreciation expense (DP _{t-1}) divided by total assets (AT _{t-1}).
Home_TACon _{t-1}	=	Home state total asset constraint for payout.
Home_StateLawIndex	=	Home state antitakeover index.
Quiet _t	=	An indicator variable coded as 1 if firm-years have negative industry- and state-adjusted growth in total assets, sales, and property, plant and equipment, and 0 otherwise.
HFIndex _t	=	An indicator variable coded as 1 if Herfindahl–Hirschman Index (HHI) is greater than annual median, and 0 otherwise. The HHI is computed as the sum of the squares of the market share of each firm in an industry each year. The market shares are computed based on firm sales and industry is defined in accordance with the Fama-French 48 industry classification.
FinDep _t	=	An indicator variable equal to 1 if industry external finance dependence is above the annual median, and 0 otherwise. Firm-level external finance dependence = (Capital expenditures (CAPX) - funds from operations (FOPT)) / capital expenditures (CAPX). When FOPT is missing, funds from operations is defined as the sum of the following variables: Income before extraordinary items (IBC), depreciation and amortization (DPC), deferred taxes (TXDC), equity in net loss / earnings (ESUBC), sale of property, plant, and equipment and investments - gain/loss (SPPIV), and funds from operations - other (FOPO). Industry level external finance dependence is the median value at the 3-digit SIC code level.
S_CashETR _t	=	State-year-adjusted one-year cash ETR estimated as the annual mean value of one-year cash ETR in the firms' state of location excluding the firm itself.
I_CashETR _t	=	Industry-year-adjusted one-year cash ETR estimated as the annual mean value of one-year cash ETR in the firm's industry excluding the firm itself.
S_CashETR _t	=	State-year-adjusted three-year cash ETR estimated as the annual mean value of three-year cash ETR in the firms' state of location excluding the firm itself.
I_CashETR _t	=	Industry-year-adjusted three-year cash ETR estimated as the annual mean value of three-year cash ETR in the firm's industry excluding the firm itself.

Figure 1. Strength of Antitakeover Law and Tax Avoidance

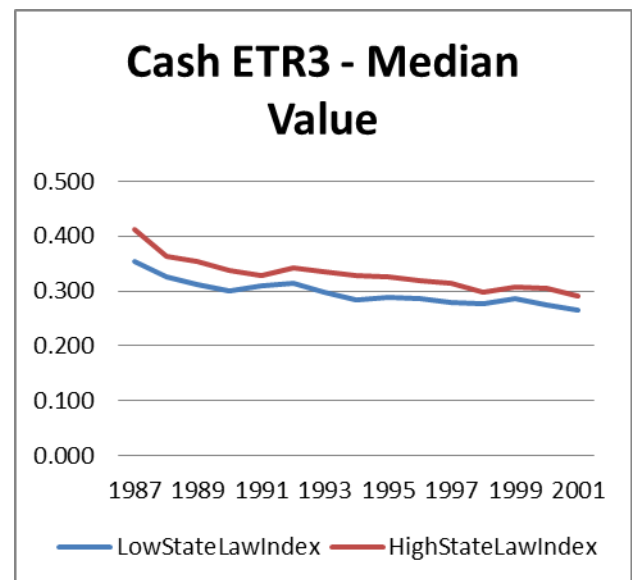
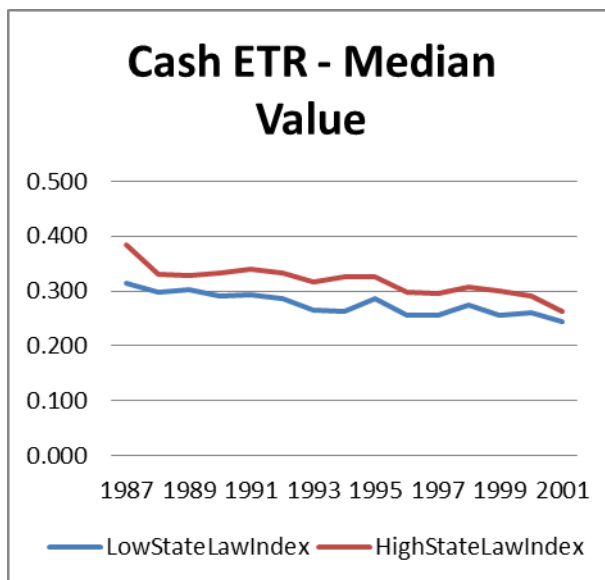
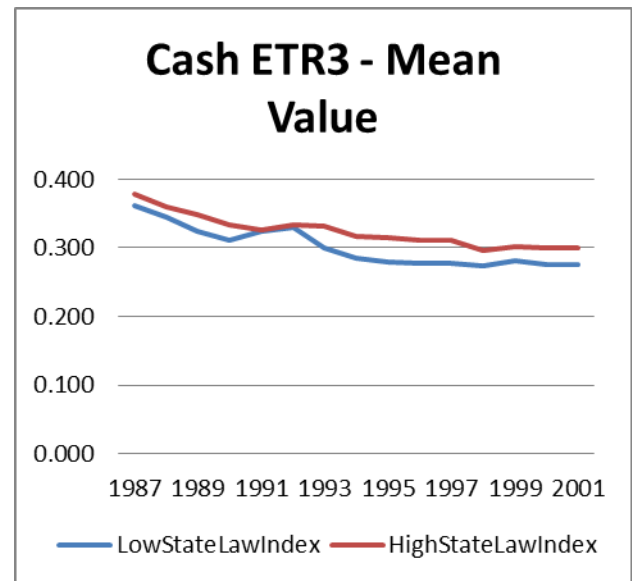
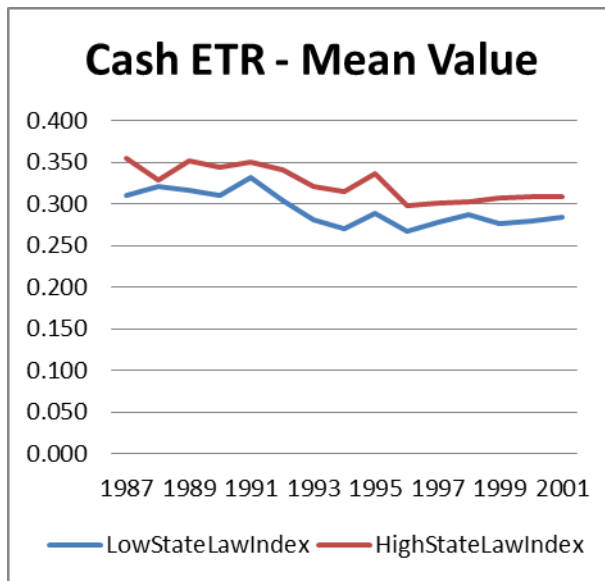


Table 1
Number of Antitakeover Statutes by State

State of Incorporation	# of Statutes	# of Firms	Percent (%)	# of Firm-years	Percent (%)
Alabama	0.00	11	0.19	41	0.18
Alaska	0.00	2	0.03	7	0.03
Arizona	4.00	18	0.31	63	0.27
Arkansas	0.00	5	0.09	36	0.16
California	0.00	228	3.96	797	3.47
Colorado	0.87	66	1.15	202	0.88
Connecticut	2.27	22	0.38	110	0.48
DC	0.00	1	0.02	4	0.02
Delaware	0.93	3,281	57.03	12,249	53.38
Florida	3.73	139	2.42	576	2.51
Georgia	3.67	89	1.55	361	1.57
Hawaii	2.73	6	0.1	29	0.13
Idaho	4.73	6	0.1	15	0.07
Illinois	3.73	33	0.57	145	0.63
Indiana	4.87	67	1.16	332	1.45
Iowa	2.07	19	0.33	107	0.47
Kansas	1.80	17	0.3	65	0.28
Kentucky	3.73	12	0.21	53	0.23
Louisiana	2.93	25	0.43	91	0.4
Maine	1.00	4	0.07	9	0.04
Maryland	3.27	52	0.9	228	0.99
Massachusetts	3.60	5	0.09	18	0.08
Michigan	2.93	73	1.27	319	1.39
Minnesota	3.73	166	2.89	741	3.23
Mississippi	2.53	5	0.09	20	0.09
Missouri	4.00	39	0.68	159	0.69
Montana	0.00	6	0.1	19	0.08
Nebraska	1.87	8	0.14	57	0.25
Nevada	4.07	136	2.36	452	1.97
New Hampshire	0.00	5	0.09	14	0.06
New Jersey	3.73	107	1.86	474	2.07
New Mexico	1.00	7	0.12	34	0.15
New York	3.87	223	3.88	1,045	4.55
North Carolina	2.80	53	0.92	248	1.08
North Dakota	0.60	2	0.03	8	0.03
Ohio	5.33	132	2.29	667	2.91
Oklahoma	1.73	19	0.33	85	0.37
Oregon	3.47	48	0.83	197	0.86
Pennsylvania	4.93	142	2.47	694	3.02
Rhode Island	3.20	9	0.16	34	0.15

South Carolina	2.80	15	0.26	91	0.4
South Dakota	4.00	5	0.09	33	0.14
Tennessee	4.60	48	0.83	213	0.93
Texas	0.33	131	2.28	501	2.18
Utah	1.87	39	0.68	157	0.68
Vermont	0.27	4	0.07	17	0.07
Virginia	3.40	91	1.58	503	2.19
Washington	2.27	54	0.94	193	0.84
West Virginia	0.00	6	0.1	30	0.13
Wisconsin	5.00	67	1.16	375	1.63
Wyoming	2.47	5	0.09	27	0.12
Total	2.48	5,753	100.000	22,945	100

This table reports the number of firms and the number of firm-years by the state of incorporation and the number of state antitakeover statutes (control share statute, fair price statute, business combination statute, poison pill endorsement statute, constituency statute, recapture statute, and staggered board statute).

Table 2
Descriptive Statistics and Correlations

Panel A: Descriptive Statistics					
Variable	Mean	Q1	Median	Q3	Std. Dev.
CashETR _t	0.299	0.129	0.286	0.399	0.230
CashETR3 _t	0.302	0.168	0.301	0.392	0.201
StateLawIndex _t	2.069	1.000	1.000	4.000	1.655
SizeAT _t	5.045	3.742	4.966	6.295	1.901
Accrual _t	0.071	0.021	0.047	0.092	0.078
ROA _t	0.124	0.049	0.102	0.172	0.115
NOL _t	0.200	0.000	0.000	0.000	0.400
ΔNOL _t	-0.006	0.000	0.000	0.000	0.052
Leverage _t	0.214	0.016	0.154	0.319	0.235
Cash _t	0.158	0.018	0.064	0.206	0.222
FIncome _t	0.008	0.000	0.000	0.000	0.024
R&D _t	0.030	0.000	0.000	0.029	0.059
PPE _t	0.359	0.156	0.287	0.496	0.272
Intangible _t	0.114	0.000	0.030	0.153	0.186
EQINC _t	0.001	0.000	0.000	0.000	0.004

Panel B: Correlations														
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
(1) CashETR _t	1													
(2) CashETR3 _t	0.615	1												
(3) StateLawIndex _t	0.057	0.051	1											
(4) SizeAT _t	0.050	0.055	-0.028	1										
(5) Accrual _t	-0.038	-0.033	-0.028	-0.222	1									
(6) ROA _t	-0.079	-0.043	-0.008	-0.097	0.176	1								
(7) NOL _t	-0.182	-0.180	-0.047	-0.091	0.059	-0.073	1							

(8) ΔNOL_t	0.122	0.115	0.009	0.171	-0.061	-0.096	-0.336	1						
(9) $Leverage_t$	-0.048	-0.059	-0.025	0.260	0.022	-0.196	0.046	0.044	1					
(10) $Cash_t$	-0.075	-0.075	-0.074	-0.215	0.156	0.401	0.013	-0.053	-0.282	1				
(11) $FIncome_t$	-0.022	-0.004	-0.054	0.219	-0.033	0.139	0.045	0.029	-0.027	0.043	1			
(12) $R\&D_t$	-0.090	-0.099	-0.082	-0.113	0.108	0.172	0.071	-0.047	-0.214	0.407	0.138	1		
(13) PPE_t	-0.107	-0.127	0.010	0.224	-0.114	-0.012	-0.050	0.057	0.355	-0.223	-0.036	-0.227	1	
(14) $Intangible_t$	0.027	0.029	-0.042	0.190	0.075	-0.021	0.029	0.036	0.379	-0.102	0.012	-0.099	-0.133	1
(15) $EQINC_t$	-0.023	-0.016	-0.007	0.128	-0.043	0.021	-0.002	0.000	0.025	-0.045	0.055	-0.043	0.033	0.019

This table presents descriptive statistics (Panel A) and Pearson correlations among the primary variables (Panel B) for a sample of 22,945 firm-years with available information on tax avoidance measures, state of incorporation, and control variables from 1987 to 2001. Correlation coefficients that are significant at the 1% or 5% level are highlighted. Variables are as defined in Appendix.

Table 3
Univariate Comparisons by Year

Year	CashETR			CashETR3		
	WeakProtection	StrongProtection	t-Stat.	WeakProtection	StrongProtection	t-Stat.
1987	0.310	0.345	-0.94	0.365	0.371	-0.16
1988	0.319	0.330	-0.71	0.346	0.362	-0.98
1989	0.315	0.353	-2.73	0.324	0.351	-2.15
1990	0.309	0.346	-2.67	0.310	0.334	-2.02
1991	0.326	0.352	-1.87	0.318	0.328	-0.85
1992	0.305	0.340	-2.82	0.328	0.338	-0.85
1993	0.281	0.322	-3.67	0.298	0.330	-2.99
1994	0.266	0.311	-4.23	0.282	0.316	-3.57
1995	0.287	0.338	-4.73	0.278	0.314	-4.35
1996	0.267	0.294	-2.75	0.276	0.306	-3.60
1997	0.276	0.295	-1.98	0.276	0.308	-3.87
1998	0.286	0.298	-1.32	0.273	0.291	-2.30
1999	0.276	0.306	-2.75	0.283	0.295	-1.41
2000	0.276	0.300	-2.07	0.274	0.297	-2.66
2001	0.281	0.301	-1.29	0.274	0.292	-1.57
Total	0.289	0.319	-9.63	0.294	0.317	-8.58

This table presents univariate comparisons in the average of one-year cash ETR (CashETR) and three-year cash ETR (CashETR3) between firms incorporated in states with weak antitakeover laws (the number of antitakeover statutes is less than 3) and firms incorporated in states with strong antitakeover laws (the number of antitakeover statutes is no less than 3).

Table 4
State Antitakeover Laws and Tax Avoidance: Baseline Regression

Panel A: OLS Regression				
Variable	CashETR_{it}		CashETR3_{it}	
	Coef.	P-value	Coef.	P-value
Intercept	0.286***	0.000	0.255***	0.000
StateLawIndex_{it}	0.005***	0.000	0.003***	0.000
SizeAT _{it}	0.007***	0.000	0.008***	0.000
Accrual _{it}	0.006	0.771	0.012	0.530
ROA _{it}	-0.126***	0.000	-0.033**	0.012
NOL _{it}	-0.087***	0.000	-0.072***	0.000
ΔNOL _{it}	0.266***	0.000	0.230***	0.000
Leverage _{it}	-0.052***	0.000	-0.049***	0.000
Cash _{it}	-0.037***	0.000	-0.045***	0.000
FIncome _{it}	-0.089	0.133	0.035	0.535
R&D _{it}	-0.310***	0.000	-0.307***	0.000
PPE _{it}	-0.096***	0.000	-0.111***	0.000
Intangible _{it}	0.027***	0.007	0.019**	0.034
EQING _{it}	-1.686***	0.000	-1.429***	0.000
Year fixed effects	Yes		Yes	
Industry fixed effects	Yes		Yes	
State of Location fixed effects	Yes		Yes	
N	22,945		22,945	
R ²	9.78%		11.58%	

Panel B: Quantile Regression				
Quantile	CashETR_{it}		CashETR3_{it}	
	Coef.	P-value	Coef.	P-value
0.1	0.006	0.000	0.009	0.000
0.2	0.007	0.000	0.007	0.000
0.3	0.007	0.000	0.007	0.000
0.4	0.006	0.000	0.005	0.000
0.5	0.005	0.000	0.003	0.000
0.6	0.004	0.000	0.002	0.000
0.7	0.004	0.000	0.002	0.006
0.8	0.001	0.267	0.001	0.781
0.9	-0.002	0.413	-0.003	0.033

This table reports the results from the OLS regression (Panel A) and quantile regression (Panel B) examining the relation between state antitakeover laws and tax avoidance. P-values are calculated based on standard errors robust to clustering at the firm-year level. ***, ** and * denote significance at the 1%, 5%, and 10% level, respectively. Industry fixed effects are based on the Fama and French (1997) 48 industry classification. The variable of interest is highlighted. Variables are as defined in Appendix.

Table 5
State Antitakeover Laws and Tax Avoidance:
Endogenous Incorporation Decisions

Panel A: First-stage regression				
Variable	HomeState_t			
	Coef.	P-value		
Intercept	0.730**	0.017		
Population _{t-1}	-0.223***	0.000		
Employee/Population _{t-1}	0.431**	0.035		
SizeAT _{t-1}	-0.027***	0.001		
CashFlow _{t-1}	0.512***	0.000		
R&D _{t-1}	0.353***	0.005		
Advertise _{t-1}	-0.218	0.692		
PPE _{t-1}	-0.003	0.900		
Depreciation _{t-1}	1.543***	0.000		
Home_TACon _{t-1}	3.061***	0.000		
Home_StateLawIndex _{t-1}	0.078***	0.000		
Year fixed effects	Yes	Yes		
Industry fixed effects	Yes	Yes		
N		21,916		
Pseudo R ²		53.74%		

Panel B: Heckman two-stage regression				
Variable	CashETR_t		CashETR3_t	
	Coef.	P-value.	Coef.	P-value
Intercept	0.191*	0.080	0.056**	0.017
StateLawIndex_t	0.005***	0.000	0.003***	0.000
SizeAT _t	0.007***	0.000	0.008***	0.000
Accrual _t	0.012	0.574	0.009	0.654
ROA _t	-0.126***	0.000	-0.034**	0.013
NOL _t	-0.088***	0.000	-0.073***	0.000
ΔNOL _t	0.285***	0.000	0.244***	0.000
Leverage _t	-0.051***	0.000	-0.049***	0.000
Cash _t	-0.038***	0.000	-0.044***	0.000
FIncome _t	-0.100	0.102	0.035	0.538
R&D _t	-0.308***	0.000	-0.307***	0.000
PPE _t	-0.101***	0.000	-0.114***	0.000
Intangible _t	0.027***	0.007	0.020**	0.027
EQING _t	-1.768***	0.000	-1.513***	0.000
InvMills _t	0.003	0.273	0.003	0.181
Year fixed effects		Yes		Yes
Industry fixed effects		Yes		Yes

State of Location fixed effects	Yes	Yes
N	21,969	21,969
R ²	9.96%	11.84%

Panel C: Covariate balance test on propensity score-matching

Variable	Homestate=1	Homestate=0	Diff.	t-Stat.
Population _{t-1}	15.889	15.812	0.077	1.75
Employee/Population _{t-1}	0.004	0.004	0.001	0.89
SizeAT _{t-1}	4.742	4.733	0.010	0.19
ROA _{t-1}	0.160	0.163	-0.003	-0.53
R&D _{t-1}	0.017	0.014	0.003	3.18
Advertise _{t-1}	0.009	0.008	0.001	1.40
PPE _{t-1}	0.366	0.401	-0.035	-0.24
Depreciation _{t-1}	0.046	0.049	-0.003	-1.13
Home_TACo _{t-1}	0.961	0.953	0.008	0.82
Home_StateLawIndex _{t-1}	3.067	3.023	0.044	0.37

Panel D: Propensity score-matching regression

Variable	CashETR _t		CashETR3 _t	
	Coef.	P-value.	Coef.	P-value
Intercept	0.493***	0.000	0.124***	0.000
StateLawIndex _t	0.006**	0.017	0.005***	0.005
SizeAT _t	0.006***	0.005	0.008***	0.000
Accrual _t	-0.018	0.687	0.013	0.743
ROA _t	-0.188***	0.000	-0.100***	0.001
NOL _t	-0.108***	0.000	-0.087***	0.000
ΔNOL _t	0.176**	0.019	0.185**	0.011
Leverage _t	-0.056***	0.003	-0.064***	0.000
Cash _t	-0.010	0.617	-0.005	0.791
FIncome _t	-0.057	0.729	0.070	0.634
R&D _t	-0.169*	0.057	-0.230***	0.002
PPE _t	-0.068***	0.001	-0.096***	0.000
Intangible _t	0.037	0.117	0.047**	0.015
EQING _t	-2.225**	0.025	-1.733***	0.008
Year fixed effects	Yes		Yes	
State of Location fixed effects	Yes		Yes	
N	4,520		4,520	
R ²	12.51%		16.60%	

Panel E: Passage of antitakeover laws

Variable	CashETR _t		CashETR3 _t	
	Coef.	P-value.	Coef.	P-value

Intercept	0.177***	0.000	0.160***	0.000
PassageBCLaw_t	0.014*	0.056	0.022***	0.001
SizeAT _t	0.005***	0.007	0.008***	0.000
Accrual _t	-0.028	0.543	0.009	0.834
ROA _t	-0.103***	0.002	0.014	0.608
NOL _t	-0.111***	0.000	-0.096***	0.000
ΔNOL _t	0.228***	0.001	0.260***	0.000
Leverage _t	-0.005	0.798	-0.016	0.336
Cash _t	-0.036**	0.029	-0.040**	0.024
FIncome _t	0.028	0.852	0.204	0.159
R&D _t	-0.338***	0.000	-0.334***	0.000
PPE _t	-0.121***	0.000	-0.127***	0.000
Intangible _t	-0.013	0.594	-0.043**	0.027
EQING _t	-0.397	0.653	0.192	0.790
S_CashETR _t	0.307***	0.000		
I_CashETR _t	0.332***	0.000		
S_CashETR3t			0.213***	0.010
I_CashETR3t			0.407***	0.000
Year fixed effects		Yes		Yes
N		4,576		4,161
R ²		11.32%		14.27%

This table reports the robustness checks on whether the relation between state antitakeover laws and tax avoidance is sensitive to the controls for endogenous incorporation decisions. Panel A report the results from the first-stage Probit regression predicting a firm's decision to incorporate in state versus out of state. Panel B presents the second stage results of the Heckman (1979) two-stage procedure after controlling for the inverse Mills ratio (InvMills) to correct for the endogenous choice of incorporation. Panel C reports the covariate balance test on the effectiveness of propensity score-matching. Panel D reports the regression results from the propensity score-matching. Panel E reports changes in tax avoidance after the passage of antitakeover laws. P-values are calculated based on standard errors robust to clustering at the firm-year level. ***, ** and * denote significance at the 1%, 5%, and 10% level, respectively. Industry fixed effects are based on the Fama and French (1997) 48 industry classification. The variable of interest is highlighted. Variables are as defined in Appendix.

Table 6
State Antitakeover Laws and Tax Avoidance: Robustness Checks

Panel A: Excluding firms incorporated in Delaware				
Variable	CashETR_{it}		CashETR3_{it}	
	Coef.	P-value	Coef.	P-value
Intercept	0.277***	0.000	0.254***	0.000
StateLawIndex_{it}	0.006***	0.006	0.006***	0.001
SizeAT _{it}	0.006***	0.000	0.007***	0.000
Accrual _{it}	-0.025	0.416	-0.027	0.324
ROA _{it}	-0.158***	0.000	-0.046**	0.016
NOL _{it}	-0.100***	0.000	-0.086***	0.000
ΔNOL _{it}	0.275***	0.000	0.266***	0.000
Leverage _{it}	-0.057***	0.000	-0.049***	0.000
Cash _{it}	-0.022*	0.068	-0.029***	0.008
FIncome _{it}	-0.086	0.405	0.010	0.911
R&D _{it}	-0.284***	0.000	-0.281***	0.000
PPE _{it}	-0.088***	0.000	-0.109***	0.000
Intangible _{it}	0.019	0.231	0.006	0.643
EQING _{it}	-1.938***	0.001	-1.621***	0.000
Year fixed effects		Yes		Yes
Industry fixed effects		Yes		Yes
State of Location fixed effects		Yes		Yes
N		10,460		10,460
R ²		11.38%		14.63%

Panel B: Alternative tax measures				
Variable	GAAPETR	CurCashETR	TBTD	ProbTaxShelter
	(1)	(2)	(3)	(4)
Intercept	0.283*** (0.000)	0.314*** (0.000)	0.028*** (0.000)	-0.606*** (0.000)
StateLawIndex_{it}	0.002*** (0.002)	0.002** (0.046)	-0.001*** (0.010)	-0.004*** (0.006)
SizeAT _{it}	0.011*** (0.000)	0.003*** (0.000)	-0.003*** (0.000)	0.197*** (0.000)
Accrual _{it}	-0.030** (0.033)	0.086*** (0.000)	0.032*** (0.000)	0.074** (0.016)
ROA _{it}	-0.033*** (0.006)	-0.139*** (0.000)	0.150*** (0.000)	1.053*** (0.000)
NOL _{it}	-0.027*** (0.000)	-0.044*** (0.000)	0.011*** (0.000)	0.012** (0.043)
ΔNOL _{it}	0.274*** (0.000)	0.325*** (0.000)	-0.196*** (0.000)	-0.353*** (0.000)

Leverage _{it}	-0.014** (0.020)	-0.058*** (0.000)	0.008*** (0.000)	-0.295*** (0.000)
Cash _{it}	-0.013** (0.020)	0.029*** (0.000)	0.010*** (0.001)	0.009 (0.471)
FIncome _{it}	-0.063 (0.163)	-1.361*** (0.000)	-0.058*** (0.003)	2.681*** (0.000)
R&D _{it}	-0.161*** (0.000)	-0.028 (0.380)	0.061*** (0.000)	-1.169*** (0.000)
PPE _{it}	-0.022*** (0.000)	-0.032*** (0.000)	0.000 (0.930)	-0.003 (0.803)
Intangible _{it}	0.109*** (0.000)	0.071*** (0.000)	-0.021*** (0.000)	0.043*** (0.004)
EQING _{it}	-2.434*** (0.000)	2.566*** (0.000)	0.531*** (0.000)	0.530 (0.370)
Year fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
State of Location fixed effects	Yes	Yes	yes	Yes
N	21,741	17,037	21,869	20,988
R ²	12.05%	12.91%	24.69%	62.97%

This table reports robustness checks on whether the relation between state antitakeover laws and tax avoidance is robust to the exclusion of firms incorporated in Delaware (Panel A) and alternative tax avoidance measures (Panel B). P-values are calculated based on standard errors robust to clustering at the firm-year level. ***, ** and * denote significance at the 1%, 5%, and 10% level, respectively. Industry fixed effects are based on the Fama and French (1997) 48 industry classification. The variable of interest is highlighted. Variables are as defined in Appendix.

Table 7
State Antitakeover Laws and Tax Avoidance: Quiet Life Hypothesis

Variable	CashETR _{it}		CashETR3 _{it}	
	(1)	(2)	(3)	(4)
Intercept	0.300*** (0.000)	0.296*** (0.000)	0.250*** (0.000)	0.265*** (0.000)
StateLawIndex _{it}	0.006*** (0.000)	0.005*** (0.001)	0.003*** (0.000)	0.003*** (0.002)
Quiet _{it} * StateLawIndex _{it}	-0.002 (0.245)		-0.000 (0.961)	
Quiet _{it}	0.011** (0.037)		0.004 (0.325)	
HHIndex _{it} * StateLawIndex _{it}		0.001 (0.495)		0.000 (0.967)
HHIndex _{it}		-0.010* (0.060)		-0.010* (0.053)
SizeAT _{it}	0.007*** (0.000)	0.007*** (0.000)	0.008*** (0.000)	0.008*** (0.000)
Accrual _{it}	0.013 (0.547)	0.006 (0.776)	0.011 (0.577)	0.012 (0.537)
ROA _{it}	-0.122*** (0.000)	-0.125*** (0.000)	-0.032** (0.019)	-0.032** (0.015)
NOL _{it}	-0.086*** (0.000)	-0.087*** (0.000)	-0.071*** (0.000)	-0.072*** (0.000)
ΔNOL _{it}	0.283*** (0.000)	0.266*** (0.000)	0.252*** (0.000)	0.229*** (0.000)
Leverage _{it}	-0.051*** (0.000)	-0.051*** (0.000)	-0.048*** (0.000)	-0.049*** (0.000)
Cash _{it}	-0.036*** (0.000)	-0.037*** (0.000)	-0.044*** (0.000)	-0.045*** (0.000)
FIncome _{it}	-0.100* (0.093)	-0.094 (0.115)	0.033 (0.554)	0.030 (0.593)
R&D _{it}	-0.327*** (0.000)	-0.309*** (0.000)	-0.323*** (0.000)	-0.305*** (0.000)
PPE _{it}	-0.099*** (0.000)	-0.097*** (0.000)	-0.114*** (0.000)	-0.111*** (0.000)
Intangible _{it}	0.025** (0.013)	0.026*** (0.009)	0.017* (0.062)	0.019** (0.039)
EQING _{it}	-1.539*** (0.000)	-1.691*** (0.000)	-1.397*** (0.000)	-1.432*** (0.000)
Year fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
State of Location fixed effects	Yes	Yes	Yes	Yes

N	22,813	22,945	22,813	22,945
R ²	9.83%	9.80%	11.72%	11.60%

This table reports the results from the OLS regression examining whether the quiet life hypothesis explains the relation between state antitakeover laws and tax avoidance. P-values are calculated based on standard errors robust to clustering at the firm-year level. ***, ** and * denote significance at the 1%, 5%, and 10% level, respectively. Industry fixed effects are based on the Fama and French (1997) 48 industry classification. The variables of interest are highlighted. Variables are as defined in Appendix.

Table 8
State Antitakeover Laws and Tax Avoidance:
Price Discount Concern Hypothesis

Variable	CashETR _{it}		CashETR3 _{it}	
	Coef.	P-value	Coef.	P-value
Intercept	0.293***	0.000	0.266***	0.000
StateLawIndex_{it}	0.004***	0.005	0.001	0.286
FinDep_{it} * StateLawIndex_{it}	0.003**	0.048	0.005***	0.001
FinDep _{it}	-0.011*	0.054	-0.017***	0.001
SizeAT _{it}	0.007***	0.000	0.008***	0.000
Accrual _{it}	0.007	0.760	0.013	0.515
ROA _{it}	-0.126***	0.000	-0.034***	0.010
NOL _{it}	-0.087***	0.000	-0.072***	0.000
ΔNOL _{it}	0.267***	0.000	0.230***	0.000
Leverage _{it}	-0.052***	0.000	-0.050***	0.000
Cash _{it}	-0.037***	0.000	-0.044***	0.000
FIncome _{it}	-0.086	0.148	0.040	0.477
R&D _{it}	-0.303***	0.000	-0.296***	0.000
PPE _{it}	-0.096***	0.000	-0.110***	0.000
Intangible _{it}	0.027***	0.008	0.019**	0.038
EQING _{it}	-1.705***	0.000	-1.458***	0.000
Year fixed effects		Yes		Yes
Industry fixed effects		Yes		Yes
State of Location fixed effects		Yes		Yes
N		22,945		22,945
R ²		9.80%		11.62%

This table reports the results from the OLS regression examining whether the price discount concern hypothesis explains the relation between state antitakeover laws and tax avoidance. P-values are calculated based on standard errors robust to clustering at the firm-year level. ***, ** and * denote significance at the 1%, 5%, and 10% level, respectively. Industry fixed effects are based on the Fama and French (1997) 48 industry classification. The variables of interest are highlighted. Variables are as defined in Appendix.